What Actually Helps a Training-Free Diffusion Predictor? A Falsification Study of Token-Adaptive Prediction on an Exact-Residual Testbed

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Abstract. Training-free diffusion accelerators such as the Token-Adaptive Predictor (TAP) reconstruct skipped network outputs from cached features, selecting per token which cheap predictor to trust by reading a first-layer *probe* and ranking predictors with a cosine proxy. We investigate three candidate improvements suggested by a single-seed pilot: an L_2 proxy in place of cosine, online least-squares predictors in place of fixed Taylor extrapolators, and the probe's per-token adaptivity. Using an exact-residual Gaussian-mixture testbed, in which the true denoiser output is available in closed form (Tweedie) and proxy fidelity is therefore directly measurable without a trained network, we conduct a 12-seed core study, a 21-regime sweep, and a pre-registered 30-test adversarial suite, all with bootstrap 95% confidence intervals. The results are largely negative. (i) The L_2 proxy improves rank fidelity in all 21 regimes, but the effect is small (ΔSpearman $+0.016$, CI $(+0.014, +0.019)$) rather than the roughly fourfold gain observed in the pilot, and increases only as the task becomes harder. (ii) Online predictors do not outperform fixed Taylor extrapolators; they are consistently worse, in 19 of 21 regimes. (iii) Probe-then-select surpasses a strong fixed predictor only under aggressive acceleration, so at moderate speedups the acceleration alone, not the adaptivity, accounts for the observed gains. We further show that per-token selection is already optimal: the objective is separable, equal-cost, and unconstrained, so greedy selection attains the global optimum. This reframes the central quantity as *proxy fidelity*, since an oracle selector retains headroom that the probe cannot capture and fidelity increases monotonically with probe richness (from 0.33 to 1.0). We also explain why an L_2 proxy tracks the L_2 output residual more faithfully than a magnitude-blind cosine.

TAP (arXiv:2603.03792) provides no public implementation; all TAP behaviour reported here rests on a faithful reimplementation. The testbed is a shallow analytic denoiser and does not include the deep learned layers that make a real first-layer probe more informative; we quantify this gap rather than leave it implicit. All code and numerical results are released.

1 Introduction

Feature-caching accelerators for diffusion transformers (DiTs) [5] skip most of the network on most sampling steps and reconstruct the missing output from cached values [8, 9]. The Token-Adaptive Predictor (TAP) generalises this idea: it maintains a family of cheap extrapolators and, for each token, selects one by reading the first-layer *probe* (the modulated input) and ranking predictors according to how closely their extrapolated probe matches the freshly computed probe under a cosine proxy. A single-seed pilot suggested three candidate improvements: replacing cosine with an L_2 proxy, replacing fixed Taylor extrapolators with online least-squares fits, and exploiting the probe's per-token adaptivity. In this work we evaluate these proposals under multi-seed, multi-regime testing to determine which of them hold.

The findings are predominantly negative, and we argue that they are more informative than an additional caching heuristic. Two of the three proposals do not survive multiple seeds, and the third yields only a small but reproducible improvement. The study also identifies the operative degree of freedom. Because TAP's per-token selection is separable, equal-cost, and unconstrained, greedy per-token selection already attains the global optimum (§5), so no selection-optimization gap remains. The binding constraint is instead the fidelity of the cheap proxy to the true, unobserved output error, a quantity that the exact testbed measures directly.

Contributions.

- An *exact-residual* Gaussian-mixture diffusion testbed (Fig. 1) in which the true denoiser output, and therefore every predictor's true error, is closed-form, making proxy fidelity measurable without a trained network.
- A proof that TAP's per-token selection has *zero* optimality gap, relocating the problem from selection optimization to proxy fidelity, together with a mechanism explaining why an L_2 proxy outperforms cosine.
- A systematic evaluation (12-seed core, 21-regime sweep, and a pre-registered 30-test suite; Fig. 2) that retires the online-predictor and probe-adaptivity proposals and bounds the L_2 improvement to a small, reproducible effect.

2 Findings at a glance

We summarise the results upfront; each is established with 12 seeds (core) or 5 seeds per cell (sweep) and 95% bootstrap CIs, and re-checked by the 30-test suite.

F1 The L_2 proxy improves on cosine in every regime, but the effect is small. Rank fidelity to the true error increases in $21/21$ regimes (ΔSpearman $+0.016$, CI $(+0.014, +0.019)$; top-1 selection accuracy $+0.028$), rather than the roughly fourfold gain reported by the single-seed pilot. L_1 performs equivalently to L_2, indicating that the operative factor is the use of magnitude information rather than cosine specifically; the advantage increases with task difficulty, from $+0.012$ at mild acceleration to $+0.034$ at cache stride 8.

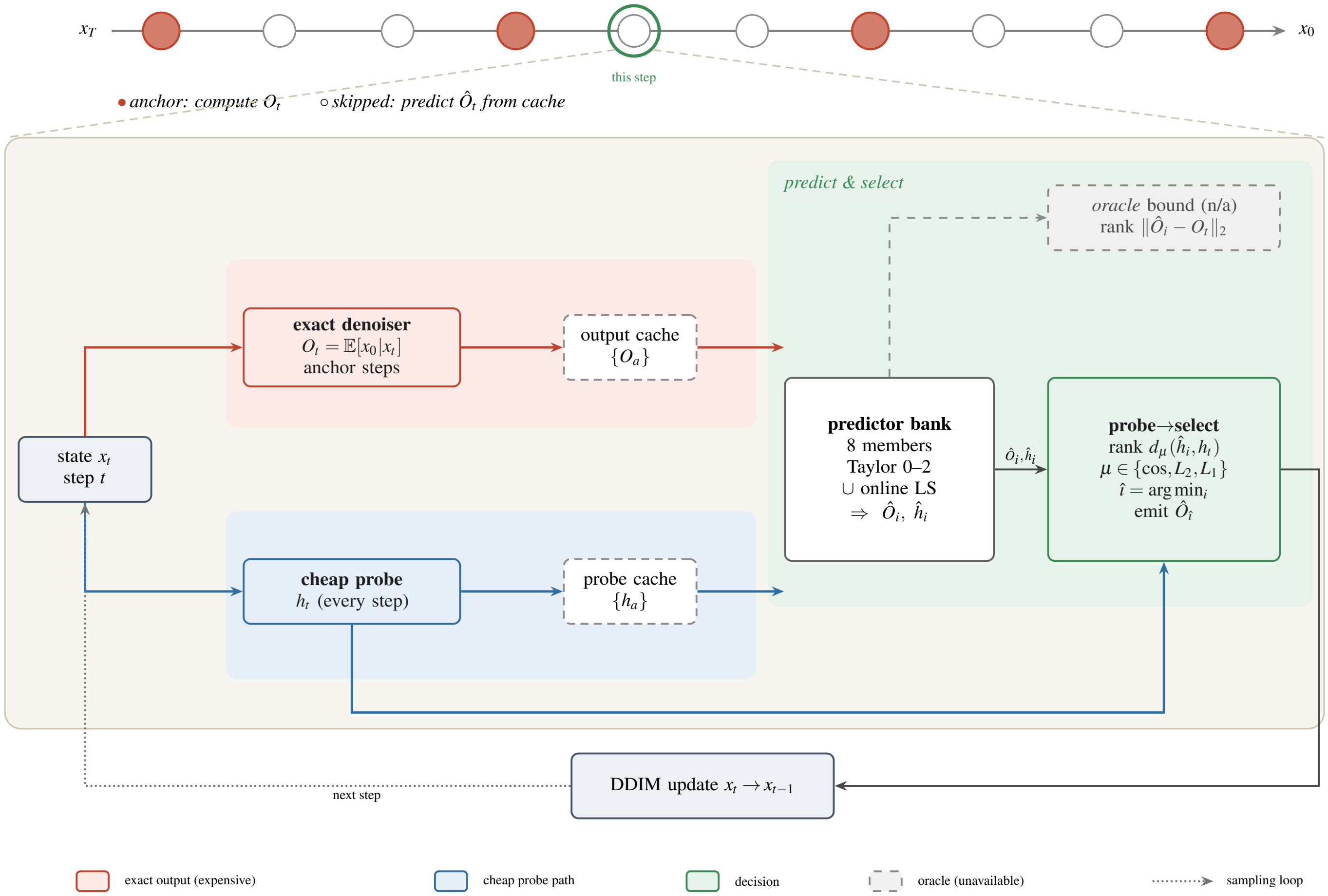

Figure 1: Architecture of the exact-residual testbed and the probe-then-select mechanism. Because the prior is a Gaussian mixture, the true denoiser output O_t (red, computationally expensive) is available in closed form, and a cheap hard-assignment *probe* h_t (blue) plays the role of a first-layer readout. At a skipped step the predictor bank extrapolates the cached outputs and probes; the selector ranks predictors by a proxy distance on the probe and applies the selected predictor's output. The *oracle* (dashed) ranks by the true output error; it is not deployable and upper-bounds the gain attainable by any proxy.

F2 **Online least-squares predictors do not outperform fixed Taylor extrapolators.** They are consistently worse (Taylor superior in $19/21$ regimes, online superior in $0/21$), and the best single fixed predictor is a Taylor extrapolator in every regime. The advantage reported in the pilot does not replicate.

F3 **The adaptivity of probe-then-select is beneficial only under aggressive acceleration.** It surpasses the best fixed predictor in exactly $1/21$ regimes (cache stride 8); in all milder regimes a single fixed predictor performs comparably, so the gains attributed to adaptive caching at moderate speedups arise from the acceleration rather than the adaptivity.

F4 **The selection problem has no optimality gap.** Per-token selection is separable, equal-cost, and unconstrained, so greedy selection provably attains the global optimum (§5). An oracle selector nonetheless outperforms the naive baseline (headroom $+0.005$, increasing to $+0.14$ at stride 8), identifying *proxy fidelity*, rather than selection or the predictor family, as the binding constraint.

F5 **Proxy fidelity is governed by probe richness.** As the probe ranges from a shallow hard readout to the exact output, fidelity increases monotonically (0.33, 0.68, 0.73, 1.0). A trained DiT's learned first-layer probe lies substantially higher on this curve (Spearman $\approx$0.77 [14]); the shallow analytic probe used here therefore lower-bounds the effect, and the measured L_2 advantage is conservative.

F6 **Methodological.** The pre-registered 30-test suite yields 26 **passes,** 0 **failures, and** 4 **qualifying notes**. Improvements that appear under a single seed diminish or reverse under multi-seed evaluation, indicating that such methods should be assessed on trained networks using paired fidelity-to-base-model metrics.

3 Background and related work

Feature caching. DeepCache [8] and FORA [9] reuse features on fixed schedules; TeaCache [10] modulates reuse by a timestep-embedding indicator; ToCa [11] and DiCache [12] make the decision token- or sample-adaptive; TaylorSeer [13] replaces cache-and-reuse with Taylor *prediction* of features. TAP generalises the predictor to a per-token *selected* member of a small family. **Diffusion samplers.** We use the variance-preserving forward process of DDPM [1] with deterministic

DDIM [2] reverse steps; the exact posterior mean follows from Tweedie's formula [6] and the score view of diffusion [3, 4]. **Evaluation hazard.** These methods are compared by generation quality at matched compute. A recurring hazard, which we make explicit and which motivated the present study, is that comparisons on analytic or untrained denoisers need not transfer to trained ones; a companion study shows that the probe's reliability is itself an emergent property of training [14]. **Status of TAP.** TAP (arXiv:2603.03792) has no released implementation at the time of writing. Every TAP-specific claim here rests on our faithful reconstruction of its probe-then-select mechanism, so we evaluate the mechanism rather than a particular codebase.

4 The exact-residual testbed

Proxy fidelity cannot be measured directly on a trained network, since computing the true skipped output would negate the acceleration it is intended to provide. We therefore adopt a prior for which the exact denoiser is available in closed form. Data are drawn from a K-component isotropic Gaussian mixture in $\mathbb{R}^d$; under the variance-preserving forward process $x_t = \sqrt{\bar{\alpha}_t}\, x_0 + \sqrt{1-\bar{\alpha}_t}\, \varepsilon$ the posterior mean is exactly

$$O_t \equiv \mathbb{E}[x_0 \mid x_t] = \sum_{k=1}^{K} r_k(x_t,t)\, \big[\mu_k + c_t\, (x_t - \sqrt{\bar{\alpha}_t}\mu_k)\big], \qquad (1)$$

with responsibilities $r_k \propto w_k\, \mathcal{N}\big(x_t; \sqrt{\bar{\alpha}_t}\mu_k, (\bar{\alpha}_t\sigma^2+1-\bar{\alpha}_t)I\big)$ and $c_t = \sqrt{\bar{\alpha}_t}\sigma^2/(\bar{\alpha}_t\sigma^2+1-\bar{\alpha}_t)$. This constitutes the computationally expensive output. The cheap *probe* h_t is the hard-assignment denoiser, retaining only the $\arg\max_k r_k$ term; it serves as the analogue of a shallow first-layer readout operating on an approximate rather than exact decision. Tokens are i.i.d. draws sharing the mixture, and DDIM steps use O at anchor steps and a predicted $\hat{O}$ at skipped steps (Fig. 1).

Predictors. A family of 8: fixed finite-difference Taylor extrapolators of order 0–2 at horizons 1–2, plus online least-squares fits (degree 2–3 over windows of 5–6 anchors). **Proxy-then-select.** Each predictor extrapolates the cached outputs ($\hat{O}_i$) and cached probes ($\hat{h}_i$); the selector picks per token $\hat{\imath} = \arg\min_i d_\mu(\hat{h}_i, h_t)$ for proxy $\mu \in \{\cos, L_2, L_1\}$ and applies $\hat{O}_{\hat{\imath}}$. **Oracle** instead minimises the true error $\|\hat{O}_i - O_t\|_2$. **Metrics.** (a) the rank correlation between proxy error and true error across the predictor family, which measures whether the proxy orders the predictors correctly; and (b) the energy distance [7] between the accelerated and full samplers' final samples, which measures end-to-end distributional fidelity. We use 12 seeds for the core study and 5 per regime cell, with bootstrap 95% confidence intervals throughout.

5 Optimality of greedy per-token selection

Let $f_n(i)$ be the true reconstruction error of predictor i on token n at a skipped step. TAP minimises $\sum_n f_n(\pi_n)$ over assignments π. Three properties hold: the objective is *separable* (token n's cost depends only on π_n), the choices are *equal-cost* (every predictor is one cheap extrapolation; no shared budget), and the feasible set is a *product* set. Hence

$$\min_{\pi} \sum_n f_n(\pi_n) \;=\; \sum_n \min_i f_n(i), \qquad (2)$$

so greedy per-token selection attains the global optimum. A regret bound or knapsack formulation over the joint assignment therefore addresses a problem with no optimality gap, which was the error in the pilot's original framing. The relevant regret is not combinatorial but statistical: the true errors $f_n(i)$ are never observed; instead we observe a proxy $g_n(i) = d_\mu(\hat{h}_i, h)$ and deploy $\hat{\pi}_n = \arg\min_i g_n(i)$. The excess error $\mathbb{E}[f_n(\hat{\pi}_n) - \min_i f_n(i)]$ is governed entirely by the rank agreement between g and f. The quantities that can reduce this excess error are a more informative metric μ or a richer probe h.

Why L_2 tracks the residual more faithfully than cosine. The true error is an L_2 residual $\|\hat{O}_i - O\|_2$, whereas cosine measures only the angle and discards magnitude, $d_{\cos} = 1 - \langle \hat{h}_i, h\rangle / (\|\hat{h}_i\|\,\|h\|)$. When predictors differ mainly in the degree to which they over- or under-shoot, which is the dominant failure mode as the extrapolation horizon grows, the discriminating signal is radial; cosine is insensitive to this component, whereas L_2 and L_1 retain it. When the probe-to-output map is locally affine near the anchors, magnitude ordering is approximately preserved, and the L_2 proxy is the better-aligned surrogate. This account predicts that the L_2 advantage should increase with acceleration and noise, both of which enlarge the radial error, a prediction the sweep confirms (§7). We present this as a mechanism rather than a theorem; the corresponding inequality holds empirically in all 21 regimes.

Table 1: Core study (12 seeds, stride-3 base regime). Energy distance to the full sampler (lower is better); paired differences with 95% bootstrap CIs.

quantity	value	paired CI
proxy Spearman, cosine	0.315	—
proxy Spearman, L_2	0.331	$+0.016\,(+.014, +.019)$
top-1 pick acc, cosine	0.393	—
top-1 pick acc, L_2	0.421	$+0.028\,(+.024, +.032)$
energy: Taylor family	0.0210	—
energy: online family	0.0249	$-0.004\,(-.006, -.002)$
energy: random select	0.0281	—
energy: naive best-fixed	0.0164	—
energy: probe-then-select	0.0207	$+0.004\,(+.002, +.006)$
energy: oracle	0.0112	$-0.005\,(-.007, -.003)$

6 Core results

Table 1 summarises the core study. The L_2 proxy improves rank fidelity and top-1 selection accuracy over cosine; the difference is statistically reliable (the confidence interval excludes zero) but modest, amounting to a 5% relative gain in Spearman correlation rather than the roughly fourfold gain suggested by the single-seed pilot. L_1 performs equivalently to L_2, confirming that the improvement derives from the use of magnitude information rather than from cosine specifically. The online-least-squares family is consistently worse than fixed Taylor, and combining the two families does not improve on Taylor alone. Probe-then-select is outperformed by the best fixed predictor in this setting ($+0.004$ in energy distance), yet the oracle selector clearly outperforms the naive baseline (-0.005): the headroom is genuine, but the probe proxy is too weak to realise it. The lim-

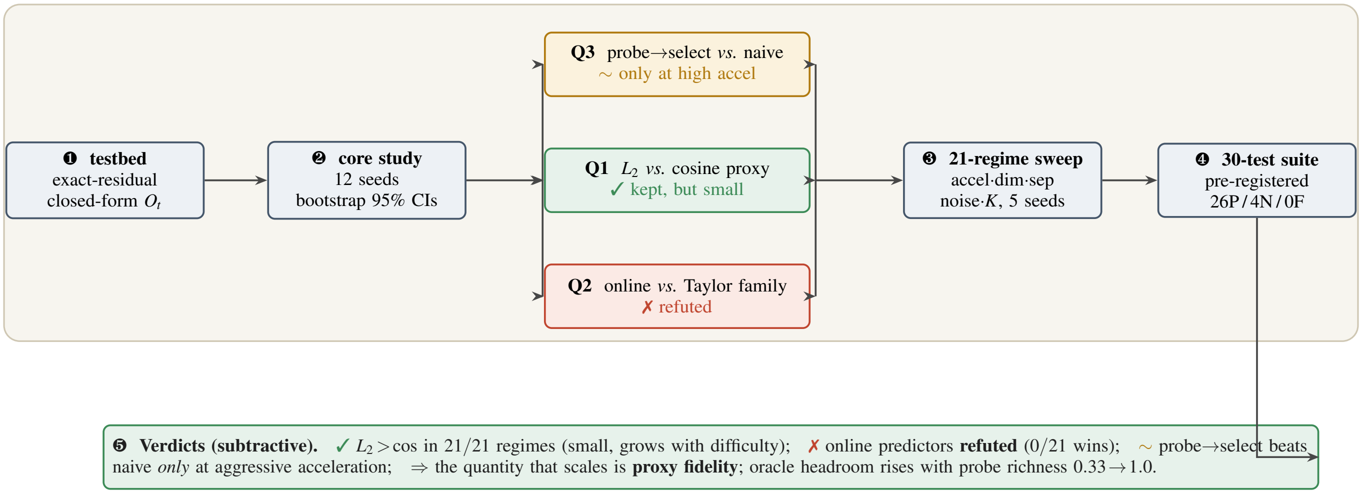

Figure 2: Experimental workflow. The exact testbed feeds a 12-seed core study that isolates three questions (Q1–Q3); each is then stress-tested across a 21-regime sweep and a pre-registered 30-test adversarial suite. Every comparison is paired and bootstrapped. The overall conclusion is negative: only the L_2 proxy holds, and it is small; the operative quantity is proxy fidelity, determined by probe richness.

iting factor is therefore proxy fidelity, not the predictor family or the selection rule.

7 Sensitivity across regimes

We vary acceleration (cache stride 2 to 8), dimension (2 to 32), mixture separation, noise scale, and component count K, giving 21 regimes with 5 seeds each (Fig. 3). **The L_2 advantage is universal but small.** It is positive and significant in all 21 regimes, increasing from $+0.012$ at mild settings to $+0.034$ at stride 8 and $+0.024$ at high noise, consistent with the radial-error mechanism. **Online predictors never win.** Taylor is significantly better in $19/21$ regimes and online in none; the best single fixed predictor is a Taylor extrapolator in every regime. **Probe adaptivity is confined to the aggressive-acceleration regime.** Probe-then-select outperforms the naive best-fixed predictor in exactly one regime, stride 8, where fixed extrapolation degrades sufficiently that per-token switching becomes worthwhile (probe$-$naive $= -0.074$). In all milder regimes the two curves coincide (Fig. 3a), so the gains of adaptive caching at moderate speedups are attributable to the acceleration rather than the adaptivity. Selection headroom increases with acceleration (Fig. 3c), so the aggressive regimes are precisely those in which a better proxy would be most valuable. **Proxy fidelity is the operative factor.** Replacing the shallow hard probe with a richer top-m soft readout increases rank fidelity monotonically, from 0.33 to 0.68, 0.73, and 1.0 (Fig. 3d). A trained DiT's learned first-layer probe lies substantially higher on this curve (our companion study measures Spearman $\approx$0.77 [14]), so the testbed lower-bounds the probe's value and the measured L_2 advantage is a conservative estimate.

8 Adversarial test suite

We pre-registered 30 tests, each designed to falsify a claim: testbed-correctness checks (exact-denoiser limits, energy(full, full)=0, and absence of true-error leakage into selection), the multi-seed core comparisons, the 21-regime crossovers, monotonicity of the acceleration–fidelity trade-off, monotonicity of a budget/abstaining selector in the compute–fidelity plane, and the probe-richness curve. The suite yields 26 **passes,** 0 **failures, and** 4 **qualifying notes**: the L_2 improvement is small rather than fourfold; the absolute proxy Spearman correlation is only moderate; the end-to-end quality gain of L_2 over cosine at the selection level is negligible; and at mild acceleration the acceleration, not the adaptivity, accounts for the measured gains. No test designed to falsify a surviving claim succeeded, and the two retired proposals failed precisely the tests intended to detect them.

9 Discussion

The results are predominantly negative. Of the three proposed improvements to a token-adaptive predictor, one (online predictors) provides no benefit, one (probe adaptivity) is beneficial only in the aggressive-acceleration regime, and one (the L_2 proxy) yields a small but reliable improvement attributable to the use of magnitude information. These observations are unified by the fact that TAP's selection is already optimal given the true errors, so the operative quantity is proxy fidelity, determined by the choice of metric and, to a greater extent, by the richness of the probe. This reframes the problem from method tuning to measurement, which the exact-residual testbed makes tractable.

10 Limitations

The denoiser is an analytic Gaussian mixture rather than a trained network, and the probe is a shallow hard-assignment readout. Both choices ensure that the ground truth is computable, but both render the probe a weaker ranker than a trained DiT's learned first layer (Fig. 3d and [14]). The absolute energy-distance magnitudes are testbed-specific, so all conclusions rely on paired, within-regime comparisons. TAP provides no public code, so we evaluate the mechanism rather than a reference implementation. Whether the small L_2 advantage and the aggressive-acceleration probe benefit increase or diminish on large latent DiTs remains open and requires computational

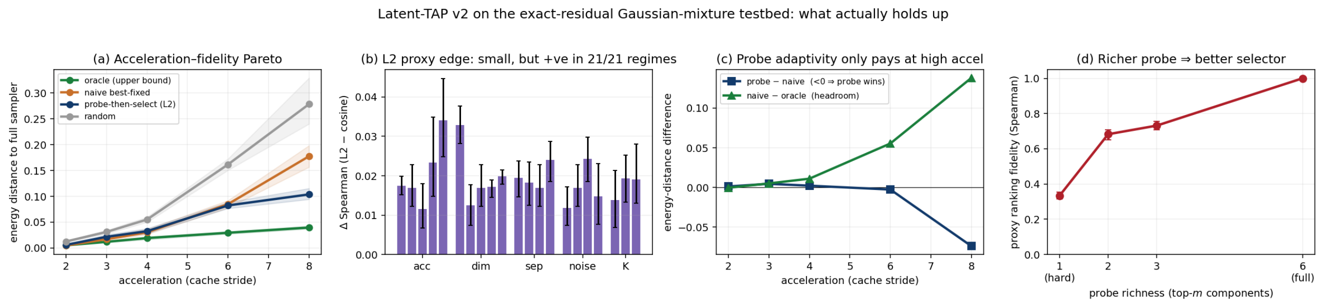

Figure 3: (a) Acceleration–fidelity trade-off: probe-then-select tracks the naive best-fixed predictor until high stride, where it separates toward the oracle. **(b)** The L_2 proxy advantage over cosine is small but positive in all 21 regimes, widening with acceleration and noise. **(c)** Probe adaptivity is beneficial only at aggressive acceleration (probe−naive crosses below 0 at stride 8), where the selection headroom (naive−oracle) is largest. **(d)** Proxy fidelity increases monotonically with probe richness, from a shallow hard probe (0.33) to the full output (1.0); this is the axis along which a trained DiT probe would improve.

resources beyond those used here.

11 Conclusion

We evaluated three proposed improvements to a training-free diffusion predictor and found that most do not hold. Under multi-seed, multi-regime testing, online predictors do not outperform fixed Taylor extrapolators, probe adaptivity is beneficial only at aggressive acceleration, and the L_2 proxy provides a small but reliable improvement over cosine because it retains the magnitude information on which the true residual depends. Since per-token selection is provably optimal, the only quantity that scales is proxy fidelity, which is determined by the metric and, to a greater extent, by the richness of the probe. We recommend that such methods be evaluated on trained networks using fidelity-to-base-model metrics, and that the probe, rather than the selection rule, be the primary target for improvement.

Reproducibility. All code and results run locally in pure NumPy: `latap_v2.py` (testbed), `run_core.py`/`agg_core.py` (12-seed core), `sweep.py`/`sweep_agg.py` (21-regime sweep), `probe_depth.py` (richness), `pressure_v2.py` (30-test suite), `plot_v2.py`; data in `results/`. See the repository README for exact commands.

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